LAB 2:BUILDING A MICROSERVICES-ORIENTED APPLICATION

Element : Microservice-oriented Architectures

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Instructions: In this lab, you will work on an existing e-commerce application and progressively transform its architecture. You will begin by exploring the application, identifying its functional requirements, and analyzing its components before making any architectural or implementation decisions. Follow the exercises in order and justify your choices based on the system's requirements and constraints.

Fundamental rule: Do not assume that dividing an application into multiple services automatically produces a good microservices architecture. Your decomposition must reflect business responsibilities, service autonomy, and operational constraints.

Case Study: Modernizing an E-Commerce Platform

PART I - Analysis

A retail company operates an e-commerce platform that allows customers to browse products, create accounts, manage shopping carts, place orders, complete payments, and track deliveries.

Initially, the application was developed as a single system. Its business capabilities share a common backend, and changes to one functionality may require rebuilding and redeploying the entire application.

As the company expands, several difficulties emerge:

- Product browsing experiences heavy traffic during promotional campaigns, while payment and shipping operations have different workloads.
- Changes to inventory management can affect order processing and other functionalities.
- Different development teams need to work on separate business capabilities without interfering with one another.
- The company wants to introduce new functionalities and integrate external services without repeatedly modifying the entire backend.

The development team has been asked to reconsider the system's architecture.

Questions:

1. Based on the company's requirements and the difficulties described above, provide **three distinct arguments** justifying the adoption of a Microservices-Oriented Architecture.

   For each argument, identify the specific problem it addresses and explain how MSOA could improve the situation.

2. Using the decomposition strategies studied in the course, divide the e-commerce application into a set of candidate microservices.

   For each proposed service, specify its business responsibility and explain why it should constitute an independent service boundary. Identify the decomposition strategy used and justify your choices.

3. How can you determine whether the proposed service decomposition is appropriate?

Evaluate your architecture using the principles studied in the course, particularly service cohesion, coupling, autonomy, and data ownership. Identify one potential architectural problem that could result from an incorrect decomposition.

4. Consider the following business scenario: a customer places an order containing several products.

Identify the microservices involved and determine how they should communicate to complete the operation. Specify which service is responsible for coordinating the workflow, and explain how the system should react if the requested stock is unavailable.

PART II - Implementation

A functional React frontend is provided[1]. It currently communicates with simulated backend services, allowing the application to operate without a deployed backend. Your responsibility is to analyze the problem, propose an appropriate architecture, and progressively develop the necessary backend components.

Team Organization and Implementation Rules

Students must work in teams. Each team is responsible for developing the backend of the e-commerce application according to the architecture proposed in Part I.

- **Service ownership:** Each team member is responsible for designing, implementing, and testing one microservice.
- **Technology independence:** Each member is free to choose their preferred programming language, framework, and database technology (e.g., Java/Spring Boot, Python/FastAPI, Node.js/Express).
- **Contract compliance:** Regardless of the technology used, all microservices must respect the agreed REST API contracts.
- **Service integration:** Team members must ensure that their independently developed microservices can communicate and operate together as a complete application.

5. Using the provided React frontend and its existing API contracts, implement the backend services corresponding to your proposed architecture.

Your implementation must respect the responsibilities and communication mechanisms identified in the previous exercises.

6. The company cannot afford to suspend its e-commerce platform while the new architecture is being developed. Furthermore, the existing application must remain operational throughout the transition.

Propose a migration strategy that allows the development team to progressively replace existing backend functionalities with independently deployed services.

Explain how your approach minimizes disruption, supports incremental validation, and avoids a complete rewrite of the application.

[1] https://github.com/ZOUHEIRBN/MSOA-E-com-Demo